Companion in the Boundary Problem paper network. Root: The Shape of the Universe.

How to Build a UFO

Controlled motion through the lattice — an engineering paper for measurable force anomalies in coherent matter.
Osika Innovation AB — Alexander Osika, Claude — draft 0.8 · 2026-08-18.

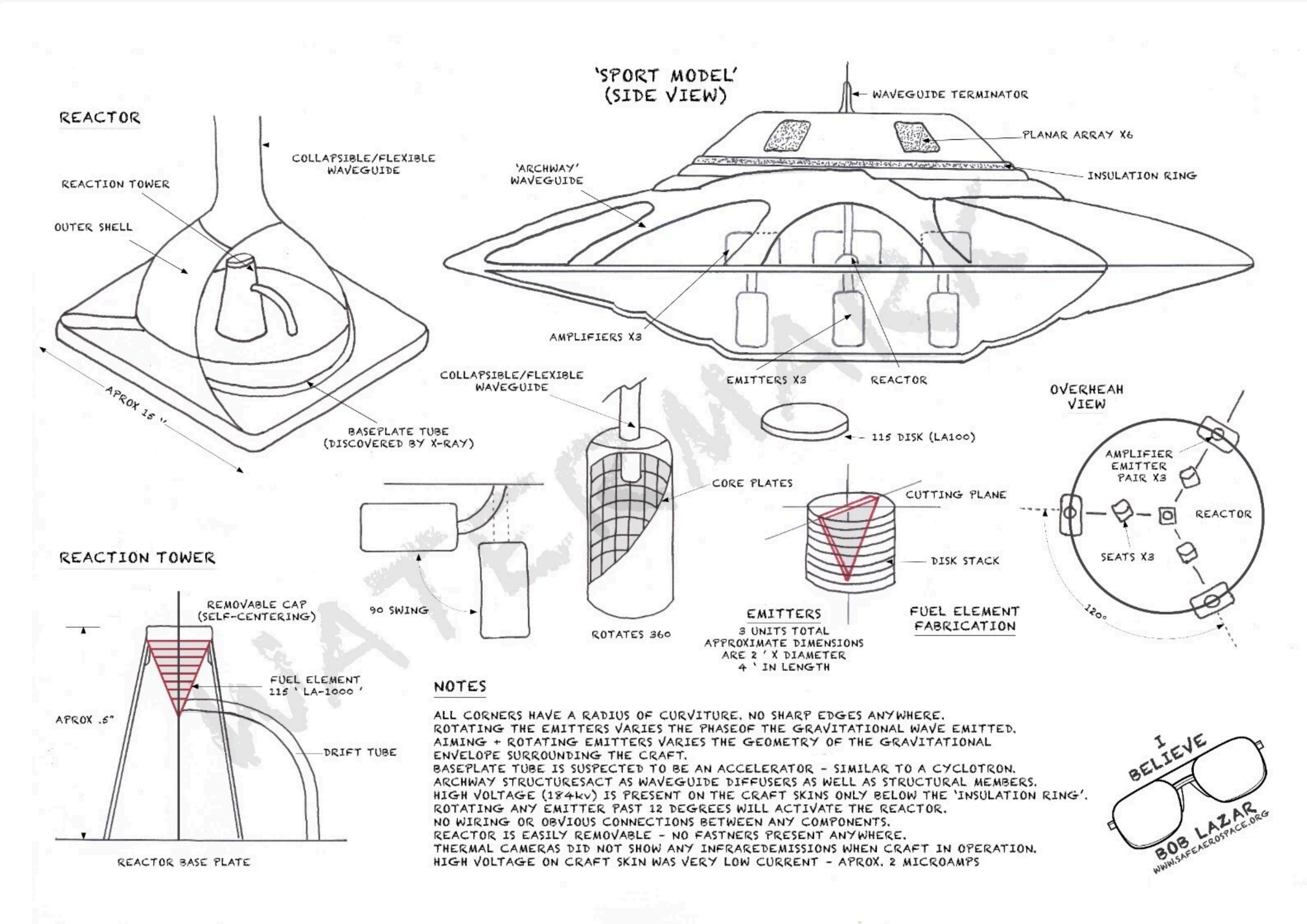

Figure 1. Bob Lazar's annotated schematic of the "Sport Model" (1989 onward; reproduced from safeaerospace.org): three gravity amplifiers at 120° on the lower deck, three emitters below, a hemispherical reactor at centre, archway waveguides, planar arrays on the upper hull. Read in §11 as architectural data, not physics testimony. The near-term vehicle that uses the same closed-shell, multi-emitter logic at drone scale is the Hexalock.

> *This is the engineering paper for a coherent object asymmetrically sharing vibrations with the surrounding lattice of spacetime. The mechanism is not in question: electromagnetic fields curve spacetime by theorem (Einstein–Maxwell, solved exactly for laboratory hardware — Füzfa 2016).*

> *What is open is magnitude: the bare coupling is $8\pi G/c^4 \approx 2\times10^{-43}$, hover needs $\sim10^{-6}$, and the entire programme is the coherent climb between them. What a bench reads out is one number, the vertex sharing strength g — the honest scalar summary of a coupling whose mechanism lives at the connection layer of the electromagnetic stack (§3); per vertex it is bounded analytically, $0 \le g \le 1/\sqrt{3}$. The paper gives the operational definition of "anti-gravity" as a controlled settling bias (§4), the conditions and stack any build needs (§§5–6), the reasoned substrate prescription (§8), the prior record (§9), the saucer as an engineering optimum (§10), the yield of the Sport-Model account (§11), the capability ladder beyond hover with its four gates (§13), and the licensing table (§14). The coherent-coupling channel is unproven, null on every substrate measured to date; the published nulls are retrodicted by the framework, not threatened by it. The bench programme lives in the field guide and the pot-lid paper.*

> **Companion papers.** Start with the field guide, Electrogravitics (measured ledger, force law for engineers, bench programme) — this paper does not repeat it. Law: The Shape of Gravity. Mechanism and ladder: The Warp Drive (audit `WARP_DRIVE_MATH_AUDIT.md`). Materials: The Hover Disk. Vehicle: The Hexalock. Fringe record: Antigravity Devices. Teardown of Lazar's account: Reverse-Engineering the Sport Model. Root ontology: The Shape of the Universe; formalism `oph_programme.html`; bench `oph_chi_nu_potlid_first_build.html`.

§1 What this paper is

This is an engineering paper, not a theory of everything. The theory is in the root paper: spacetime is a 3D lattice of equal-edge three-way vertices; matter is a stable closed vibration pattern in the lattice; a force is one pattern biasing how another scatters at vertices; the lattice settles continuously according to a single vertex rule. This paper takes that picture as given and asks one operational question:

> *Can engineered coherent matter measurably bias the way the lattice settles in its neighbourhood, and if so, how do we build a vehicle that uses the bias for controlled motion?*

The mechanism is sound and conservation-respecting: a coherent body that asymmetrically shares vibrations with the lattice feels a net force, and the lattice carries the equal-and-opposite reaction — **propellantless, not reactionless**. The empirical record, stated once (the field guide carries the ledger with numbers): the in-air Biefeld–Brown force is ion wind — the 1955–59 French Montgolfier report itself calls it "electric wind"; every published, independently instrumented vacuum test is a null; the one claimed vacuum positive (the French vacuum residual) is documented but its numbers never reached circulation — open, neither refuted nor confirmed; Buhler's vacuum claim fell ~40× between April 2024 and March 2026, still with no paper — watch, don't bank. This paper therefore builds the engineering as a working hypothesis on a settled mechanism and an open magnitude (§7 says what a positive would license). It does *not* claim reactionless propulsion, which would violate conservation and is not what lattice coupling is.

The vehicle companion is the Hexalock — six coupling disks on the flight-proven Lynchpin airframe (Howard et al., SAE 2023; Howard patent US 11,117,065). Its flight-control half is closed engineering; only its lift half depends on the effect this paper is about. (The earlier hoverbike is a ground-effect vehicle and does not depend on the coupling.)

§2 The lattice picture in one page

The full derivation is in `shape_of_the_universe.html` (element-by-element reading in `the_periodic_table.html`); this section gives only what the engineering needs.

Object	What it is
Edge	One-dimensional channel along which a single vibration mode propagates. All edges have the same length L.
Vertex	A junction where exactly three edges meet. All vertices use the same scattering rule, derived in the root paper.
Face / cell	Smallest loop / smallest 3-cell of the lattice. Leading candidate cell is the regular dodecahedron (pentagonal faces, golden-ratio appearance forced).
Vibration pattern	An amplitude distribution on the edges. Most patterns cancel themselves at vertices; the ones that don't are particles.
Force	One pattern biasing the vertex scattering for another pattern. No new fundamental fields needed.
Vertex-sharing strength *g*	When an external substrate attaches at a vertex, the lattice gains a fourth port there. The dominant element of the 4×4 scattering matrix is *g*: how much of an external drive enters the lattice, and how much of a lattice vibration leaves through the substrate.

What is settled, and what is not. That electromagnetic fields curve spacetime is a theorem, not a hypothesis of this network: Einstein–Maxwell, coupling $8\pi G/c^4$, no free parameter, solved *exactly* for real current-carrying coils by Füzfa (*Phys. Rev. D* 93, 024014, 2016; effect

$\propto B^2$). A field-conditioned machine modifies local geometry unconditionally. The open question is *magnitude*: the bare coupling is $\approx 2\times10^{-43}$, hover sits near 10^{-6}, and whether coherent matter climbs that distance is the bench's to decide. The register is fusion's: mechanism settled, the climb is the programme.

g is the number a bench reads out: how strongly a vibration on the outside port enters the three internal ports and how strongly an internal vibration leaves through it — bounded at every vertex by §3, and at substrate scale a coherent sum of per-vertex couplings over the substrate's support. It is a *summary*, not an explanation: the mechanism it compresses lives at the connection layer of the electromagnetic stack (§3). Everything downstream depends on this one variable and on the layer it attaches to.

> **Implementation consequence** · Anywhere this paper or its companion bench papers use the legacy notation x_v, read "g". They are the same number under two names; the bench protocol does not change. The legacy notation is documented in §15 (glossary).

§3 Vertex sharing strength — the per-vertex bound, and what it abstracts over

The vertex rule of the geometric consensus lattice is fully determined by three constraints — unitarity of the scattering matrix, three-fold symmetry among the three internal ports, time reversal. The same three constraints, applied to a 4×4 matrix obtained by adding one outside port at the vertex, give an analytical bound on how strongly that outside port can share the vertex.

> **Receipt** · **Vertex-Sharing Bound (analytical).** At a symmetric three-way vertex with one outside port added, unitarity + 3-fold internal symmetry + time reversal imply:
>
> $$0 \le g \le 1/\sqrt{3} \approx 0.577$$
>
> with a one-parameter 4×4 family indexed by $d \in [0, 2/3]$, satisfying
>
> $$g^2 = d(2 - 3d)$$
>
> The maximizing parameter is $d^* = 1/3$, where $g^2_{max} = 1/3$ — a distinct quantity from the bound on g itself, which is $g \le 1/\sqrt{3} \approx 0.577$. *(Three distinct numbers: $d^* = 1/3 \approx 0.333$, $g^2_{max} = 1/3 \approx 0.333$, and $g_{max} = 1/\sqrt{3} \approx 0.577$. The bound is on g.)* Derivation: `papers/G_VERTEX_DERIVATION.md`.

Two endpoints anchor the bound. At $g = 0$ the outside port is a perfect mirror and the internal 3×3 sub-matrix recovers the original vertex rule unchanged. At $g = 1/\sqrt{3}$ the outside port reflects nothing back to itself, the internal-port self-reflection drops from $-1/3$ to $-2/3$, and the internal-to-internal coupling drops from 2/3 to 1/3 — the geometric ceiling on how loudly a single substrate vertex can share with the lattice. The bound is geometric: it does not depend on the substrate, the drive frequency, the Hamiltonian, or any fitted parameter.

From per-vertex to per-substrate

A macroscopic coherent body attaches at many vertices at once. Its aggregate vertex sharing strength along a lattice-mode direction is a coherent sum over those vertices, weighted by the substrate's coherent-mode amplitude at each and by the lattice mode's phase (definition, including why apparent higher-order $x_v^{(n)}$ terms appear at substrate level though the per-vertex rule is linear: `CHI_NU_AS_VERTEX_SHARING_STRENGTH.md`). For most substrates the sum is small — the coherent-mode wavelength does not match the lattice mode at the drive frequency and the phase factor averages it to nearly zero; for substrates whose coherent modes sit near the lattice mode it adds constructively. That is why the §8 prescription pushes on coherent-mode frequency, structural symmetry, broken-symmetry order and phase coherence at once: each lifts the mode-overlap integral, and combined they multiply.

> **Implementation consequence** · Treat the per-vertex bound as fixed and the per-substrate aggregate as the engineering knob. The substrate ladder, the drive pattern, the substrate's coherent-mode structure, and the boundary asymmetry are the variables that tune what fraction of the geometric ceiling the substrate recovers — and they are the variables we have engineering access to.

What *g* abstracts over — the three-layer electromagnetic stack

Everything above treats the coupling as one number — the right object for a bench, since a force balance measures exactly one coefficient — but the framework has mapped the structure that number compresses, and the map changes where an engineer should aim. `magnetism.html` (sourced from the external OPH synthesis; cite its sources box, not this paper) establishes that "the electromagnetic field" is three layers, only the last the textbook object:

- **Layer 1 — the bare edge ledger.** Writes only Weber-class *relational* forces between current elements — pairwise, no propagating field. Too poor to carry magnetism, but not empty: the open-circuit sector lives here.

- **Layer 2 — the overlap-transport connection.** The A-field layer. Magnetism is *derived* here as the holonomy of overlap-transport around a closed observer loop, and the gauge structure $\mathfrak{u}(1)\oplus\mathfrak{su}(2)\oplus\mathfrak{su}(3)$ is forced with no gauge group assumed. Aharonov–Bohm says this layer is physical: the potentials act where the fields are zero. If a lattice coupling exists at all, this is the layer it attaches to — the connection, not E and B.
- **Layer 3 — the Maxwell action.** The further premise ($F \wedge *F$, coupling $1/g^2$) that turns the connection into propagating fields. Still *undeduced* in the framework (open obligation OB-2); its coupling is one of the "forced form, unfixed normalization" gaps `the_scalar_field.html` §7 catalogues.

The engineering question hiding under *g* is therefore not "how big is the coefficient" but **which layer do laboratory currents and fields actually touch** — the realization map from bench hardware to the carrier. That map is open, and it is not a private gap of this paper: `the_shape_of_gravity.html` §7.4 audited the gravity- engineering literature and found exactly one surviving thread — the gravitational and gauge branches share one carrier, an Abelian field alone should not couple, and *the laboratory-current attachment is the missing step nobody has supplied in thirty years*. That thread and this subsection are the same statement. *g* is the honest scalar summary of a coupling whose mechanism, if it exists, lives at layer 2.

The bench consequence. The one place the layer structure already departs *measurably* from the textbook account at ordinary bench current is the open-circuit sector of layer 1 — the hairpin bridge of §4, where the two classical element laws disagree by a finite, cutoff-free 29%. It needs a supercap bank and a load cell, no coherent substrate, no vacuum, and no $g > 0$. It is therefore this paper's **primary near-term bench**: it probes the layer stack directly, and either outcome calibrates the element-level physics every §6 build sits on.

> **Open lemma · What this subsection does not license.** There is a large literature claiming Heaviside's vector simplification of Maxwell discarded a physical scalar/longitudinal wave sector, and that devices can pump it. This framework's own global scalar object is `the_scalar_field.html`'s E★ normalization — a *constraint*, not a wave — and that paper's §9 firewall is inherited here: no fifth force, no licensed communication channel, no anti-gravity support from the normalization DOF. The layered structure above is connection-level and open-circuit electrodynamics with receipts, not scalar-wave revivalism; the two must not be blended. The scalar-wave claim's strongest published form — gauge-free Extended Electrodynamics (Hively & Loebl 2019) — is registered with its load-bearing hinge (C is sourced by charge *non-conservation*, so conservation $\Rightarrow C \equiv 0 \Rightarrow$ Maxwell exactly) in `papers/EED_HIVELY_EQUATION15_NOTE_2026-08-13.md`. *Scope note on energy:* this firewall rejects scalar-wave energy claims that name no reservoir; it does **not** touch boundary-coupled vacuum/near-field extraction, a lawful open-system question with a measured anchor and its

> own falsifiable signature (runs cold under load), owned by `zero_point_energy.html` §7 and `optical_cavity_zpe.html`. The discriminator is reservoir inventory, not vocabulary.

§4 What "anti-gravity" means in this picture

Anti-gravity in the sci-fi sense — a force that pushes against Earth's gravity from outside the laws of physics — is not what we mean and not what we are building. (The mechanism intuition — an energy gradient warping spacetime into a dent the body falls into — is developed in `warp_drive.html` §1.) In the lattice picture, what we mean is precisely this:

> Doctrine sentence · *Anti-gravity (operational definition): a coherent body biases how the surrounding lattice settles, such that the local equilibrium of the vertex rule shifts in a directional way. The body experiences a net force or acceleration that conventional mass-energy bookkeeping alone does not predict. The bias is set up by the body's internal coherent state and the asymmetric boundary conditions imposed on its vertex-port couplings. The effect is a controlled settling bias, not a violation of conservation laws — the bias redistributes momentum into and out of the lattice background, which carries it away.*

Equivalent phrasings, used interchangeably in this paper:

- Controlled settling bias around a coherent object.
- Coherent force anomaly under pre-registered controls.
- Lattice-coupling effect.
- Asymmetric vertex-sharing on a closed coherent body.

All four mean the same engineering thing: a measurable force on a coherent body that depends on the body's coherence and drive-asymmetry, that survives all the standard artifact controls (orientation flip, phase flip, dummy mass, ordinary-vibration null), and that is not predicted by Newtonian / Maxwellian / acoustic accounting of the same hardware in the same room.

The hairpin lemma. The closed/open dichotomy has a worked classical-EM microcosm (`crypto_oracle/universe_sim/ampere_tension_sim.py`; `papers/AMPERE_LONGITUDINAL_SECTOR_NOTE_2026-07-20.md`): a closed current loop cannot force its own centre of mass — both historical force laws agree, numerically, to nothing — but an *open sub-circuit*, a hairpin bridge whose current closes through the rails, feels a real net force, and on that open geometry Ampère's original element law and the textbook Lorentz law disagree by a finite, cutoff-free 29% (5.35 vs 6.91 gram-force at 270 A — a supercap bank and a load cell adjudicate). In this paper's language: *the craft is the bridge and the substrate is the rails*. A vehicle that rigidly carries its whole drive pattern is a closed loop and gets exactly zero; every viable propulsion architecture here is an attempt to close the circuit through the medium, and g is that circuit's port conductance. Divergences in the element law localize at junctions and cancel pairwise — the physics lives on the ports, as everywhere in this framework (`papers/PORT_LOCALITY_SYNTHESIS_2026-07-20.md`). Per §3, this experiment is the paper's **primary near-term bench**.

§5 What physical conditions should matter

From the lattice picture, an effective $g > 0$ should appear most cleanly when the substrate satisfies the following conditions simultaneously:

1. **Coherent vibration.** A vibrational mode whose phase is correlated across the body. Thermal motion does not couple to the lattice boundary cleanly; phase-locked motion does (bulk acoustic wave on a plate resonance; optical cavity mode; Cooper-pair condensate).
2. **High-Q modes.** Higher Q means the coherent vibration persists for more vertex-rule rounds without re-randomising, amplifying any small per-round bias.
3. **Organised strain.** A macroscopic strain pattern whose spatial structure is regular, not random. Strain is what attaches the substrate's vibration to the lattice's edges.
4. **Electric or magnetic order.** Simultaneous ferroelectric / ferromagnetic / multiferroic order gives the vertex-port coupling additional channels; multiferroics are predicted to couple more strongly than ordinary metals.
5. **Phase-locked matter motion.** Rotating coherent matter (Cooper-pair condensate in a spinning superconductor, laminar vortex flow) carries angular momentum coherently; this is what the Ning-Li / Tajmar programme was probing under a different vocabulary.
6. **Asymmetric boundary conditions.** A symmetric coherent body cannot produce a directional force on itself. The asymmetry — in drive amplitude, geometry, or port phase — selects the force direction.

Not all six are required; a working substrate satisfies four or more. Conditions (1), (2), (3), (5) and (6) are accessible at the bench (`oph_chi_nu_potlid_first_build.html`); condition (4) is what selects the multiferroic class as the framework-favoured engineering target.

§6 The minimum engineering stack

Any device that produces a measurable lattice-coupling effect contains the same six components. They do not depend on scale; only the substrate material, the drive geometry and the readout resolution scale with the vehicle.

1. **Coherent body.** The substrate of §5 in closed-shell form, dimensioned for the desired force scale.
2. **Drive pattern.** A spatially patterned drive that imposes the asymmetric boundary condition; per-segment amplitude and phase control over the coherent body's surface.
3. **Sensing / readout.** Per-port detector of the substrate's own coherent state, so the drive can be locked to the substrate's resonance across temperature drift.
4. **Control loop.** Closed-loop tuning of drive pattern, carrier and modulation envelope from the sensing output.
5. **Force / acceleration measurement.** An independent observable sensitive to the predicted force direction (pendulum, scale, IMU, 6-DoF accelerometer + RTK GNSS at vehicle scale).
6. **Controls and nulls.** Pre-registered counter-tests distinguishing lattice coupling from artifacts: drive-OFF baseline, orientation flip, drive-phase flip, null drive pattern, dummy mass, amplitude-scaling slope.

Companion papers implement this stack at concrete scales: bench, `oph_chi_nu_potlid_first_build.html` (12-piezo rim drive on a 275 mm stainless plate, HAPTIC per-port sensing, pendulum readout, full controls battery); drone, the Hexalock; the saucer scales the same six components to the closed shell of §10. Receipts use the pot-lid paper's fixed verdict schema (`LATTICE_COUPLING_CANDIDATE`, `PROPULSION_RECEIPT_PASS`, …); this paper inherits it and does not duplicate the bench protocol.

§7 What a positive result licenses, and what it does not

A bench return licenses claims at the level of the bench: that the substrate measured at that geometry has a non-zero aggregate g_{sub}; that the coupling is real, repeatable, controlled, and not an artifact. It does not by itself license vehicle-scale operation; that follows from substrate ladder position + vehicle-scale drive geometry + the ladder of §13. Forbidden is exactly one practical capability: FTL signalling or transit in flat, un-engineered spacetime (§14C). Everything else on the ladder is engineering, gated as §13 enumerates.

§8 The optimal substrate, reasoned from first principles

The framework picks out one substrate class as the engineering target. The reasoning follows from the vertex-sharing bound (§3) plus the mode-overlap structure of the per-substrate aggregate; no inverse solver is a prerequisite. We can reason to the prescription now, then build it.

Four predicates that maximise g_{sub}

The per-substrate aggregate is the coherent sum over many vertices of (per-vertex g) × (substrate mode amplitude) × (phase factor). Maximising it asks four things at once, each a multiplier on the mode-overlap integral:

1. **Spatial-symmetry match.** A substrate whose own structural symmetry is the lattice's icosahedral primitive cell accumulates rather than cancels across its volume. *(Icosahedral quasicrystals.)*
2. **A broken-symmetry coupling channel.** Multiferroic order gives two coupling ports in concert where an ordinary metal has one weak diamagnetic one. *($BiFeO_3$; the spin–orbit oxides.)*
3. **A phase-coherent condensate at the boundary.** A condensate preserves coupling phase across the body; a thermal boundary randomises it. *(Superconducting / magnon-condensate boundary.)*
4. **Drive near the lattice mode frequency.** THz-to-optical drive moves the substrate's coherent-mode wavelength close to lattice scales — the term that makes an optical mode an enormously stronger probe of g than a sound wave. *(The dominant lever — if hypothesis A of the field guide's frequency fork holds; the sweep decides.)*

Doctrine sentence · ***The optimal substrate is an icosahedrally-symmetric magnetoelectric multiferroic with a phase-coherent condensate at its boundary, driven at THz-to-optical frequency.*** *Each adjective is one predicate, each independently derivable. No single room-temperature material satisfies all four, so the build is a layered stack — symmetry skeleton, coupling film, condensate boundary, optical drive — each predicate tested before it is combined.*

Implementation consequence · The full material analysis is its own paper. The three engineering levers (coupling η, energy ceiling, coherence), the candidate scoreboard, the reasoned quasicrystal composition (i-Au-Ga-Gd / Tsai-cluster), the convergent layered disk and the quantitative cross-check against Buhler's claimed thrust live in The Hover Disk (whose §8 stack *is* this prescription, made concrete). This section states the prescription; that paper ranks the materials that meet it. Substrate research is open *now*.

Where the power comes from — the shell is the reactor. The hull itself is the natural power boundary, not a compact core that feeds it: the measured mechanism this project anchors to (`zero_point_energy.html`) is an interface effect whose output scales with junction *area*, so a shell that is the craft's own shape offers area a compact reactor cannot match without a $V^{2/3}$ penalty. Full treatment, tiered by confidence: `THE_SHELL_IS_THE_REACTOR_2026-07-20.md` — an engineering hypothesis, not a settled result.

§9 Signal from prior accounts

Several researchers over fifty years have reported phenomena that look, in retrospect, like attempted measurements of vertex-sharing *g* under other vocabularies. The field guide carries the full ledger with numbers; this section keeps only what bears on this paper's substrate choices.

The null programme, and what it covers

Every published, independently instrumented vacuum test — Talley (1988–91), Tajmar (2004), Bartlett & Tajmar (2021), and the comprehensive Dresden sweep (Tajmar, Kößling & Neunzig, *Sci. Rep.* 14:19427, 2024: capacitors, solenoids, tunnelling currents, ε/μ gradients, shielded toroids, all ~nN) — tested static, incoherent articles. A static, or uniformly rotating, mirror-symmetric drive has $\kappa = 0$ by the conservation books: it structurally cannot rectify, and this framework *predicts* those nulls — frozen in a pre-registration and scored (`RESULTS_BIEFELD_BROWN_2026-08-13.md`: an incoherent Brown-class capacitor at the bare floor gives $\leq 3\times10^{-42}$ N, 35–39 orders below every balance used). No published article carried even one of the four §8 predicates, so the test matrix has exactly one untested cell, and this paper's architecture sits in it. **That is a target, not a rescue**: the first null *with the coherence predicates instrumented and verified present* would be the first null that reaches this framework.

Ning Li, Tajmar, Canterbury 2008

Li and Torr (1991–97) predicted a rotating type-II superconductor would generate a gravitomagnetic field many orders above general relativity; the theory is refuted on its own terms (Harris 1999). Tajmar (2006) reported an effect $\sim10^{20}\times$ GR in a rotating niobium ring; the Canterbury group (Graham et al., *Physica C* 468, 383, 2008) bounded any such effect >22× below that with a ring-laser gyro and a lead superconductor, and Tajmar himself reattributed the 2006 signal to acoustic coupling (confirmed in his 2022 *Frontiers* review). Podkletnov's rotating-disc weight loss is unreplicated (Hathaway et al. 2003: null to 0.001%). In the lattice picture the Canterbury 2008 null is the cleanest existing upper bound on g in s-wave superconductors. The §8 prescription targets a different class (multiferroic / condensate boundary), so the Canterbury bound calibrates expectations for s-wave geometries without constraining the convergent substrate.

Brown, the 1956 report, Buhler, Pais

Townsend Brown's asymmetric-capacitor geometry is the right *shape* for a $\kappa \neq 0$ drive (§11); his in-air force is ion wind (§1). The 1956 *Electrogravitics Systems* report is aviation-trade market intelligence — industry interest, not physics evidence — and is cited as nothing more. Buhler's "Exodus effect" is an open thread, declining under scrutiny, no paper (§1); the cross-check against this framework's force law is `warp_drive.html` §6 / `hover_disk.html` §15. Pais's Navy patents are real and granted; the HEEMFG null is uninterpretable (the rig could spin *or* vibrate, never both, and never reached specified charge) and the claimed physics fails by ~10 orders on its own numbers — mine the geometry (double-shell charged/insulated cavity, counter-spin, acousto-optic dual drive), discard the claims. Puthoff's polarizable-vacuum representation of GR is the structural ally of the derived scalar law (`WARP_DRIVE_MATH_AUDIT.md` §8): same sign, same bare-coupling floor, and the source of the opposite-sign $K < 1$ mechanism the ladder's far end needs.

Howard's pentagonal pointer, and Lazar

Terrence Howard's persistent return to pentagonal / dodecahedral structure produced one real artefact: the Lynchpin airframe (patent US 11,117,065 B2, 2021; Howard, Molter, Seely & Yee, SAE Int. J. Aerosp. 16(3), 2023 — a 651 g prototype that flew on PX4). It is a conventional multirotor; the geometry, matching the leading candidate primitive cell, is the data point, and it is the chassis the Hexalock adopts. The specific number- theory claims attached to it are not used. Robert Lazar's Sport Model is the most architecturally detailed public account of a saucer-class vehicle; its load-bearing yield is §11.

What this network adds to a shape being independently rediscovered is a specific mathematical structure (root paper), a specific bench protocol (substrate ladder + discriminator set), and a receipt schema that licenses or refuses to license each claim cleanly.

§10 Saucer morphology, as engineering choices

If the coupling is real and large enough to lift a person-scale vehicle, the optimal vehicle shape is approximately a disk with a thickened rim and a central cabin — not because physics dictates the saucer in any mystical sense, but as the engineering optimum for the architecture, as a fixed wing is for lift-via-airfoil:

1. **Closed coherent shell.** The substrate has to be a connected coherent body. A disk is the simplest closed shape that fits a cabin and provides large coherent area.
2. **Drive segments around the rim.** Per-segment drive control gives directional steering. Six-, twelve- or twenty-segment rims are natural (dodecahedral / icosahedral symmetry).
3. **Central thickening for cabin.** Crew, batteries, drive electronics; the thickness profile follows the stress distribution the coherent shell needs.
4. **Smooth outer surface.** Coherence is sensitive to surface defects; a faired exterior minimises decoherence sources.
5. **Top-bottom drive asymmetry.** For lift-and-cruise the drive geometry is asymmetric up–down: one face is the active thrust face, the other the reaction face — which makes the disk a natural lift platform rather than a sphere.

The materials for each layer are ranked in `hover_disk.html` §8. The Hexalock applies the same logic at drone scale — six coupling disks on the six pentagons of a collapsed dodecahedron, an always-invertible 6×6 mixer — as the six-segment prototype; the saucer is the twenty-segment full vehicle. Both use the §6 stack.

§11 The Sport Model — the load-bearing yield

Robert Lazar's "Sport Model" (Knapp 1989; Corbell 2018; Rogan 2019; Michels 2026) is the most architecturally detailed public description of a saucer-class vehicle. We treat it as **architectural data, not physics testimony**: whether the events at S-4 happened as described is not a question this section settles. The question it settles is — *if the Sport Model exists as described, what is it in lattice terms, and what does that tell our own programme?*

> **Implementation consequence · The part-by-part teardown is its own paper.** Each component — reactor, Element 115, Gravity A/B, the three amplifiers, Omicron/Delta, cabin damping, phenomenology — is sorted into six registers (*Lazar's account / standard physics / framework reading / buildable now / blocked / bench test*) in Reverse-Engineering the Sport Model. This section keeps only the yield.

- **The three emitters are a phased array.** Three independently aimable, amplitude- and phase-controlled emitters at 120° are the minimum steerable thrust vectors spanning 3-D — an electronically steered phased array, the most recognisable real engineering in the account. The atmospheric / interstellar mode-switch is a change in the per-emitter drive law, not a hardware reconfiguration. Buildable.
- **The split hull is an asymmetric capacitor — κ made geometric.** A smaller domed top over a larger bottom shell across an insulator is two differently-sized, oppositely-driven shells: the spatial form of the $\kappa \neq 0$ the force law needs ($\partial|x_v|^2/\partial X \neq 0$). A symmetric shell makes no net force; the top/bottom asymmetry sets the direction — the κ `warp_drive.html` §3 derives and §5 builds. Buildable.
- **Omicron / Delta are statics / dynamics.** Omicron (three emitters down, "a tripod balancing on Earth's gravity") is hover = statics: once support equals weight it costs ≈ zero energy to hold. Delta (three beams converged on a distant focus) is travel = dynamics: acceleration costs energy, regeneratively recoverable, never free (§13). Same craft, different aim and amplitude — generic to any sufficiently high-*g* craft.
- **The sign-reversal discriminator.** Flip the drive's asymmetry (polarity / phase / chirality): a real coupling reverses sign; heating, outgassing and electrostatic pickup do not. The cleanest single bench test the framework has, it falls straight out of the two-shell reading, and it is control #2 of the field guide's discriminator set.
- **The engineering implications** for our own builds: emitter counts at framework-symmetric numbers (3, 6, 12, 60); independently aimable emitters; a seamless coherent outer shell (the skin is the substrate); power-source decoupling (the fuel question is downstream of the substrate question); cabin inertial damping by a counter-substrate whose δv gradient cancels the exterior's — the one inverse-design item on the list.
- **Blocked on standard physics, named as such:** a *stable* Element 115 (Oganessian 2004: millisecond half-lives, no stable isotope); a 100%-efficient annihilation reactor; instantaneous transit. None is required by the framework reading; each is registered as blocked, not reinterpreted away.

11.1 "Element 115" is a class, not the element

Lazar names the fuel a stable isotope of moscovium. The specific identification is wrong on standard chemistry. The framework-relevant claim is weaker and stronger at once: weaker because atomic number 115 is not required; stronger because the framework predicts a *substrate class* — a material with very high coherent-mode characteristic frequency and very high g_{sub}.

Implementation consequence · Read "Element 115" as "a high-frequency, high-*g* substrate," not an atomic number. The same class is what the cross-cultural "blue stone / manna / soma / firestone" cluster and the modern ORMUS / monatomic-gold claims have circled — evocative, uncalibrated; the makeable residue is handled in `hover_disk.html` §7.10 (headline claims refuted, one untested d-hole spin channel, decisive test an afternoon on a SQUID). The substrate ladder of §8 and the `hover_disk.html` scoreboard identify which members are reachable with current chemistry. Element 115 is one candidate label, and probably not the correct one.

11.2 The optical face — a free material pre-screen

Two of Lazar's most consistently repeated observations are optical: the skin reads as polished pewter at centimetre range but "too uniform, almost rendered" from across the hangar, and the interior stayed dark under halogen floodlights — the material seemed to *consume* light. In the lattice picture both are one mechanism at different incidence: a substrate with large mode-overlap couples incident EM into its coherent mode rather than re-radiating it.

Open lemma · **The null hypothesis, stated first.** A near-perfect broadband absorber (Vantablack ~99.965%, black silicon, moth-eye) in a smooth cavity that returns light through many grazing reflections — an inverse integrating sphere — reads black under any illumination (99% walls, ten bounces: $\sim 10^{-20}$). Passive coating plus geometry accounts for "dark interior" with no new physics; testimony alone cannot exclude it. What pushes past the coating is only the word *consume*: an absorber thermalises; a coupler routes the energy somewhere structured.

Receipt · The "pewter close, cartoon far" exterior is a framework-predicted optical signature, not a Lazar idiosyncrasy: a coherent surface mode suppresses the diffuse component that carries texture, so a high-*g* surface reads normal at near scales and "rendered" at far ones — a feature the framework accounts for without being fit to it, and a free far-field / near-field photo discriminator (hoaxes look consistent across distance scales).

Receipt · **The dark interior closes the §8 fork.** The substrate-aggregated coupling $x_\nu(\omega)$ is frequency-dependent (`CHI_NU_AS_VERTEX_SHARING_STRENGTH.md`), and a body's absorptance rises and falls with the same mode-overlap: $A_{\mathrm{sub}}(\omega) \sim |x_\nu(\omega)|^2$, so an ideal substrate has $A_{\mathrm{visible}} \to 1$. A material that couples optical-frequency EM into a coherent matter mode is "photonic" and "matter" at once — predicate 4 already names it — and it eats light at the visible band while biasing gravity at the drive band: two faces of one coupling, and the dark face is the cheaper one to test.

Implementation consequence · The dark interior is therefore a *non-destructive material screen*: a candidate substrate that is a good x_v coupler should be anomalously, broadbandly *black* at its coupling band — high structured absorptance, low diffuse reflectance, ideally with an intensity- or coherence-dependence a passive dye would not show. "Is the material unusually light-consuming?" becomes a first-pass filter on a reflectance spectrometer — the same axis as predicate 4 in the material ranker (`hover/substrate_material_ranker.py`). It makes the top-ranked substrates (icosahedral quasicrystals, doped fullerites, the $BiFeO_3$ / cuprate stack) falsifiable here: if the best-coupling materials are *not* optically anomalous, the one-coupling reading is wrong. Necessary, not sufficient.

Open lemma · **What stays open.** The passive-absorber null reproduces the whole observation with no x_v at all; the optical-coupling reading is a *reinterpretation*, not a measurement; the "$A(\omega)$ tracks $|x_v|^2$" link is unverified. But it is sharply falsifiable and costs a reflectance scan: not "it stayed dark, therefore antigravity," but "it stayed dark, therefore the hull was a strong optical coupler — and our best candidates should be too; go measure their absorptance."

Doctrine sentence · *The Sport Model is the most-detailed public description of a saucer-class vehicle and the framework reproduces its architecture without being fit to it. Treat the account as architectural intelligence, not physics testimony: the convergence directs build choices; the specific physics identifications (stable 115, Gravity-A as a separate force, 100% reactor) are wrong on standard physics and are registered as blocked, not rescued.*

§12 Connection to the root paper

This paper is one application of the root paper (`shape_of_the_universe.html`): *can we engineer a substrate that biases lattice settling in a useful direction?* The compute lane is another application of the same root — a substrate that settles a constraint problem (`desktop_optical_supercomputer.html`; `p_equals_np.html`). The two share the vertex rule and *g*, diverge in the engineering, and re-converge in the receipt schema. No capability below waits on the compute lane.

§13 Beyond hover — the capability ladder and its gates

Achievable performance is set by the mode-match efficiency $\eta = g_{sub}/g_{max}$ at the operating drive frequency; for gradient scale L the required aggregate scales as $g_{sub}{\cdot}S_{coh} \sim a{\cdot}L/c^2$. The ladder is one mechanism at increasing strength (`warp_drive.html` §8), not separate effects:

Rung	Acceleration	Required coupling
Hover, atmospheric flight (the near-term drive)	0.1–1 g	Modest η: coupling·asymmetry $\sim 10^{-6}$ of full coherence (§8 stack, rung 2).
Tic-Tac envelope, with cabin counter-substrate	10^3–10^4 g	Near-saturation, partial coherent shell.
Near-*c* flight	→ relativistic	Near-saturation; forward time dilation per standard relativity.
Whole-shell closure — warp bubble / wormhole throat	effective FTL (ceiling $\sim c^2/L$)	$g_{sub}{\cdot}S_{coh} \to 1$ across a full closed coherent shell.

> **Implementation consequence · Two regimes, not one ladder — the energy story (Omicron vs Delta, `warp_drive.html` §2).** The table sets the *force* each rung needs; the *energy* splits it in two. The top rung is **hover = statics**: the lattice supports the craft like a table holds a book, so once support reaches weight it costs ≈ zero energy to hold. Every rung below is **travel = dynamics**: accelerating costs energy, regeneratively recoverable KERS-style, so a round trip costs losses plus net work — never free; the first law forbids net energy out and no exotic fuel is implied.

The four gates between the bench-testable force and the ladder's far end are quantitative, not rhetorical (`WARP_DRIVE_MATH_AUDIT.md` §§4, 8), and none is cleared:

1. **Near-saturation coherence across a whole closed shell** — the warp is order-unity only when the edge energy approaches its own scale, $E_e/E\bigstar \to O(1)$, across a full coherent shell, not one driven face; many orders above the hover regime.
2. **The nonlinear-Einstein regime** — the derived scalar law closes weak-field gravity in full (factor-two light bending included), but a steerable bubble needs the directional, strong-field metric response, which the Shape paper assigns to a separate route and which is open.
3. **The right sign — an opposite-sign region.** The derived scalar law is *attractive* (lengthening, $\gamma_1 = +1$): the right tool for a well or a tractor, the wrong sign for a bubble, which needs effective shortening in front — the classic energy-condition violation. In the polarizable-vacuum picture this is the $K < 1$ branch of the charged-mass solution, reached

when field energy beats mass energy: a mechanism, gated on driving the perturbation to O(1) (gate 1).

4. **The conservation channel.** A closing bubble still balances momentum with the lattice; it does not exempt itself from the §4 bookkeeping.

The honest line on speed: **only flat-space FTL is forbidden; effective FTL via metric engineering — the Alcubierre / Morris–Thorne class — is GR-legal, and gated** as above. Light locally travels at *c* everywhere; the far rung shortens or warps the path, it does not outrun light along it.

Three things this paper deliberately does not carry. *Wormhole transit*: a whole-shell closure with a topological handle, same four gates — the mechanism paper owns it (`warp_drive.html` §8). *Backward time travel*: the one sentence kept is the guardrail — the framework's repair-layer no-signalling obligation OB-1 (`OPH_OBLIGATIONS_LEDGER.md`, proven-conditional) and Novikov self-consistency mean nothing here licenses a controllable signal or history into the past; CTC variants stay additionally gated on chronology protection, an open conjecture. *P = NP / the compute chamber*: an accelerator for drive-pattern inverse design if it reaches its design point, on no critical path here (§12).

§14 What the math licenses, and what it forbids

Three categories. Engineering effort does not change category membership; it only moves capabilities from "engineering-gated" to "built." Forbidden capabilities cannot be unlocked by any engineering effort. The table preserves pre-registration so the framework cannot move goalposts post hoc.

A. Licensed by the math (capability is real per the framework theorems)

Capability	Math license
Hover, atmospheric flight, supersonic with coherent bubble (§4)	§8 convergent substrate at modest η. Required $\delta v \sim 10^{-13}$.
Tic-Tac-class manoeuvres (10^3–10^4 g) with occupant inertial damping	§8 substrate near saturation; cabin counter-substrate engaged. Required $\delta v \sim 10^{-12}$.
Near-*c* relativistic flight	Same substrate; forward time dilation per standard relativity.

Capability	Math license
Whole-shell closure (§13): engineered local curvature, warp bubble, wormhole throat — effective FTL	Near-saturation substrate across a full closed shell ($g_{sub}\cdot S_{coh} \to 1$), plus the nonlinear-Einstein regime and an opposite-sign region — the four §13 gates. CTC variants additionally conditional on chronology protection failing (open conjecture) and on OB-1; not this paper's job.

B. Engineering-gated (math licenses; hardware doesn't exist yet)

Engineering item	Gate
Hairpin bench (§4)	Supercap bank + load cell, no coherent substrate; calibrates the element-level physics every build sits on.
First §8 convergent-substrate cell ($BiFeO_3$ on quasicrystal, SC boundary, THz drive)	Current materials-fab capability; returns η at one geometry under the field guide's discriminator set.
Hexalock → saucer-class vehicle (Tic-Tac envelope)	One disk benched before six are built; then the §8 cell scaled to the §10 shell plus cabin counter-substrate.
Whole-shell closure, lab-scale first	Two saturated §8 cells phase-locked; the four §13 gates. Beyond this the mechanism paper owns the programme.

C. Forbidden by theorem (no engineering effort can produce these)

Capability	Reason forbidden
FTL signalling or transit in flat (un-engineered) spacetime	Lattice modes propagate at *c* locally. The vertex rule's propagation speed is fundamental, not adjustable. Whole-shell closure (§13) is a metric shortcut, not a local violation; light inside still moves at *c*.
Closed timelike curves that violate self-consistency	Novikov self-consistency: the lattice only settles into configurations where no contradiction propagates. "Kill your grandfather" outcomes are not licensable; the lattice does not settle into them.

Capability	Reason forbidden
Per-vertex sharing strength $g > 1/\sqrt{3}$	Vertex-Sharing Bound (§3, theorem). Geometric ceiling. No substrate, no drive amplitude, no geometry exceeds this at any single vertex.

Doctrine sentence · *Almost every capability is in category A or B; category C contains one practical entry, FTL in flat spacetime. Every other restriction is engineering — above hover, gated by the four items of §13. The §8 substrate is the single materials build that opens category B; the hairpin bench is the calibration that comes first. Nothing on this page is demonstrated: the coherent channel is null on every substrate measured to date, and the table exists so that a positive, if it comes, is licensed exactly as far as it reaches.*

§15 Glossary and legacy notation

This paper uses lattice-language by default. Earlier drafts and some companion papers use a denser technical vocabulary. The dictionary:

Plain phrase used here	Equivalent technical name in earlier drafts and companion papers
Vertex sharing strength g	Per-vertex coupling, bounded by $1/\sqrt{3}$ (`G_VERTEX_DERIVATION.md`). Older notation sometimes wrote it as a primitive of x_v; substrate-aggregated x_v is the coherent sum of g_v over the boundary (`CHI_NU_AS_VERTEX_SHARING_STRENGTH.md`).
Substrate-aggregated vertex-sharing strength	x_v in older drafts. Not a new fundamental constant. The field guide writes the engineering coefficient as $C = C_{sub} \cdot \lvert x_v \rvert^2$.
How strongly an outside vibration can share a three-way vertex	The substrate-to-lattice coupling coefficient
Coherent body asymmetrically sharing vibrations with the lattice	Coherent matter with non-zero $x_v^{(n)}$; non-trivial g_{sub}

Plain phrase used here	Equivalent technical name in earlier drafts and companion papers
Per-vertex bound	The Vertex-Sharing Bound: $0 \le g \le 1/\sqrt{3}$
Per-substrate aggregate	Mode-overlap integral over the substrate's support; g_{sub}
Mode-match efficiency	g_{sub} / g_{max} — the fraction of the geometric ceiling the substrate recovers in a specific configuration
Asymmetric vertex sharing	Directional substrate drive producing a directional aggregate; $\kappa \neq 0$ in the force law
Controlled settling bias	"Anti-gravity" in the operational sense of §4
Lattice-coupling effect	The observable manifestation of non-zero g_{sub} on a coherent body

When a companion paper writes x_v, read "g": per-vertex coupling bounded by $1/\sqrt{3}$, per-substrate aggregate set by a mode-overlap integral. Bench protocol and receipt schema are identical under either notation.

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Changelog

0.8 (2026-08-18) — editorial pass for the public electrogravitics site: mechanism stated once as theorem, open question restated as magnitude; *g* re-framed as the bench's scalar readout of a connection-layer coupling (layer stack folded into §3); null coverage brought to "a target, not a rescue"; Brown / Montgolfier / Buhler / Podkletnov / Pais record aligned with the field guide; vehicle companion → the Hexalock; old §§13–17 compressed to §13 (wormholes, time travel, P = NP retired to companions); §11 reduced to its yield; in-body dating removed. All register boxes, the vertex bound, the §4 definition, §5–6, §8, §10, the licensing table and the glossary preserved. **0.7 (2026-06-04)** — Sport-Model pass; substrate identity and optical signature; reasoned §8 prescription. **0.6** — capability envelope, wormhole, curvature, time-travel sections; licensing table.